

Observing with Mathematical Perspectives

A Textbook for Math in Society

MATH& 107 — Centralia College

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Chapter 1

Number Systems

This chapter reviews the main number systems used in mathematics and explains how they relate to one another. We begin with the natural numbers and then move through the integers, rational numbers, and real numbers. Each system includes the ones before it, while also allowing us to describe a wider range of situations.

The chapter also reviews the basic operations on real numbers and introduces two common recursive patterns. These ideas provide part of the numerical foundation for the topics that follow later in the course.

1.1 The Number Tower

One useful way to organize the number systems is as a sequence of enlargements. We start with numbers used for counting, then extend to numbers used for subtraction, division, and measurement. This layered view helps explain why the standard notation

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

is more than just a list: it records a progression of increasingly flexible number systems.

Definition. *Natural numbers.* The **natural numbers**, denoted $\mathbb{N}$, are the counting numbers $\{1, 2, 3, 4, \dots\}$.

Definition. *Integers.* The **integers**, denoted $\mathbb{Z}$, extend the natural numbers by including zero and the negative whole numbers: $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

Definition. *Rational numbers.* The **rational numbers**, denoted $\mathbb{Q}$, consist of all numbers expressible as a fraction $\frac{a}{b}$ where a and b are integers and $b \neq 0$.

Definition. *Real numbers.* The **real numbers**, denoted $\mathbb{R}$, include all rational numbers together with all irrational numbers. The irrational numbers are the real numbers that cannot be written as a fraction of integers.

The containment is strict at every level: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Every natural number is an integer. Every integer is a rational number. Every rational number is a real number. Not every real number is rational.

$\mathbb{R}$ Real numbers every position on the number line

$\mathbb{Q}$ Rational numbers all fractions a/b with $b \neq 0$

$\mathbb{Z}$ Integers $\dots, -2, -1, 0, 1, 2, \dots$

$\mathbb{N}$ Natural numbers $1, 2, 3, \dots$

Example 1.1.1. (*Classifying numbers*) Classify each of the following numbers by placing it in the smallest system that contains it.

- (a) 7 is a natural number (and therefore also an integer, rational, and real).
- (b) -3 is an integer that is not a natural number.
- (c) $\frac{2}{5}$ is a rational number that is not an integer.
- (d) $\sqrt{2} \approx 1.41421\dots$ is a real number that is not rational. Its decimal expansion continues without repeating.
- (e) $\pi \approx 3.14159\dots$ is a real number that is not rational.

Exercises 1.1

1.1.1. Classify each number as natural, integer, rational, or real (using the smallest applicable category).

- (a) 12
- (b) -5
- (c) $\frac{3}{8}$
- (d) $\sqrt{5}$
- (e) 0
- (f) $\frac{6}{3}$

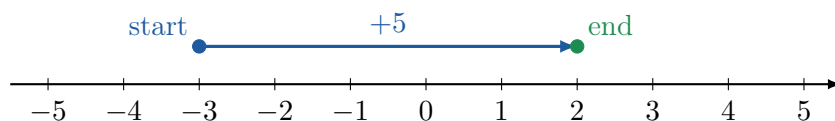
1.1.2. True or false. Justify each answer in one sentence.

- (a) Every integer is a rational number.
- (b) Every rational number is an integer.
- (c) The number $0.333\ldots = \frac{1}{3}$ is rational.
- (d) There exist real numbers that are not rational.

1.1.3. The decimal 0.75 terminates. Write it as a fraction in simplest form and identify which number system it belongs to. Explain why $\sqrt{3}$ cannot be written this way.

1.2 The Additive View: Numbers as Steps

Addition and subtraction are often easiest to understand on the number line. In that setting, a number tells us how far to move and in which direction. Positive values move to the right, negative values move to the left, and zero leaves the starting position unchanged. Subtraction can then be interpreted as adding the opposite number.



This interpretation is useful because it connects arithmetic to motion. Instead of memorizing separate rules, we can think about where a calculation starts and how the operation changes that position.

Example 1.2.1. Compute $-3 + 5$. Starting at -3 , move 5 units to the right.

$$-3 + 5 = 2.$$

Example 1.2.2. Compute $4 + (-7)$. Starting at 4, move 7 units to the left.

$$4 + (-7) = -3.$$

Example 1.2.3. Compute $4 - (-7)$. Subtracting -7 is the same as adding 7.

$$4 - (-7) = 4 + 7 = 11.$$

Example 1.2.4. (*Fractional steps*) The rational numbers allow steps of any size. Starting at $\frac{1}{3}$, move $\frac{1}{2}$ unit to the right.

$$\frac{1}{3} + \frac{1}{2} = \frac{2}{6} + \frac{3}{6} = \frac{5}{6}.$$

Exercises 1.2

1.2.1. Compute each of the following.

- (a) $-8 + 5$
- (b) $-2 + (-6)$
- (c) $7 - (-4)$
- (d) $0 + (-9)$
- (e) $\frac{1}{4} + \frac{1}{4}$
- (f) $\frac{2}{3} + \frac{1}{6}$

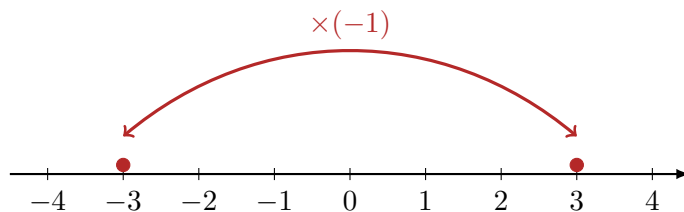
1.2.2. A hiker starts at position -6 on a trail map. The hiker walks 10 units forward, then 14 units backward.

- (a) Write each move as an addition and give the position after each move.
- (b) Mark all three positions on a number line.

1.2.3. A checking account has a balance of $-\$230$ (overdrawn). A deposit of $\$500$ is made, followed by a withdrawal of $\$185$. Write the final balance as a sequence of additions and give the result.

1.3 The Multiplicative View: Numbers as Scalings

Where addition is naturally described as movement, multiplication is naturally described as scaling. Multiplying by a positive number changes a quantity by a factor. If the factor is greater than 1, the distance from zero increases. If the factor is between 0 and 1, the distance from zero decreases. A negative factor also reverses direction across zero.



This point of view also helps explain the sign rules. Multiplying by -1 reflects a number across zero. Doing that twice returns to the original side, which is why the product of two negative numbers is positive.

Example 1.3.1. $3 \times 4 = 12$. The distance from zero is stretched by a factor of 4.

Example 1.3.2. $3 \times \frac{1}{4} = \frac{3}{4}$. The distance from zero is reduced to one-quarter.

Example 1.3.3. $(-2) \times (-5) = 10$. Two reversals produce a positive result.

Example 1.3.4. (*Division as reverse multiplication*) Division by b is multiplication by $\frac{1}{b}$.

$$12 \div 4 = 12 \times \frac{1}{4} = 3.$$

Dividing by a fraction smaller than 1 produces a result larger than the original.

$$6 \div \frac{1}{3} = 6 \times 3 = 18.$$

There are 18 thirds in 6.

Exercises 1.3

1.3.1. Compute each of the following.

(a) $(-3) \times 7$

(b) $(-4) \times (-5)$

(c) $8 \times \frac{1}{4}$

(d) $10 \div \frac{1}{5}$

(e) $\frac{2}{3} \times \frac{3}{4}$

(f) $\frac{5}{6} \div \frac{1}{3}$

1.3.2. Explain in one complete sentence why $(-1) \times (-1) = 1$, using the idea of reversal.

1.3.3. Without computing, state whether each result is greater than or less than the original number. Justify each answer.

(a) $7 \times \frac{3}{2}$

(b) $7 \times \frac{1}{2}$

(c) $7 \div \frac{1}{2}$

(d) $7 \div 3$

1.4 Rational Numbers and Equivalence

Rational numbers are important because they allow us to describe parts of a whole, ratios, rates, and many forms of comparison. A single rational number can be written in many different ways as a fraction. For example, $\frac{1}{2}$, $\frac{2}{4}$, $\frac{3}{6}$, and $\frac{50}{100}$ all represent the same value. Understanding why these expressions are equivalent is an important part of working with fractions.

Definition. *Equivalent fractions.* Two fractions $\frac{a}{b}$ and $\frac{c}{d}$ are **equivalent** when $a \times d = b \times c$.

Example 1.4.1. Are $\frac{3}{4}$ and $\frac{9}{12}$ equivalent?

$$3 \times 12 = 36 \quad \text{and} \quad 4 \times 9 = 36.$$

The products are equal, so the fractions are equivalent.

Example 1.4.2. Are $\frac{2}{5}$ and $\frac{7}{15}$ equivalent?

$$2 \times 15 = 30 \quad \text{and} \quad 5 \times 7 = 35.$$

The products differ, so the fractions name different locations.

Simplest Form

Although many fraction representations may describe the same number, it is often useful to write a fraction in simplest form. A fraction is in **simplest form** when the numerator and denominator have no common factor greater than 1. In practice, this means dividing both by their greatest common factor.

Example 1.4.3. Simplify $\frac{18}{24}$. The greatest common factor of 18 and 24 is 6.

$$\frac{18}{24} = \frac{18 \div 6}{24 \div 6} = \frac{3}{4}.$$

Irrational Numbers

The number $\sqrt{2}$ is the positive real number whose square is 2. It cannot be expressed as a fraction of integers. Its decimal expansion is 1.41421356..., which continues without repeating. Numbers of this kind are called **irrational**. Other familiar irrational numbers include π , $\sqrt{3}$, and $\sqrt{5}$.

Exercises 1.4

1.4.1. Determine whether each pair of fractions is equivalent. Show your calculation.

- (a) $\frac{4}{6}$ and $\frac{6}{9}$
- (b) $\frac{5}{8}$ and $\frac{15}{25}$
- (c) $\frac{7}{11}$ and $\frac{21}{33}$
- (d) $\frac{3}{7}$ and $\frac{9}{20}$

1.4.2. Write each fraction in simplest form.

- (a) $\frac{12}{16}$
- (b) $\frac{15}{35}$
- (c) $\frac{24}{36}$
- (d) $\frac{100}{250}$

1.4.3. The decimal 0.6 can be written as $\frac{6}{10}$. Write it in simplest form, then verify using the equivalence test that your simplified fraction names the same location.

1.4.4. Explain in one complete sentence why $\sqrt{4}$ is rational but $\sqrt{2}$ is not.

1.5 Operations on Real Numbers

The real numbers support the main arithmetic operations used throughout the course. Some of these operations combine two numbers, while others undo or reverse a previous operation. Reviewing them together helps clarify the role each one plays.

Operation	Notation	Description	Example
Addition	$a + b$	Move right by b	$3 + 5 = 8$
Subtraction	$a - b$	Move left by b	$8 - 5 = 3$
Multiplication	$a \times b$	Scale by b	$3 \times 5 = 15$
Division	$a \div b$	Scale by $\frac{1}{b}$	$15 \div 5 = 3$
Squaring	a^2	Multiply a by itself	$4^2 = 16$
Square root	$\sqrt{a}$	Reverse squaring	$\sqrt{16} = 4$

Squaring and taking square roots are inverse operations on nonnegative real numbers. For that reason, they are often used together when solving equations and interpreting measurements.

Example 1.5.1. Compute each of the following.

$$7 + (-3) = 4$$

$$7 - (-3) = 10$$

$$6 \times \frac{1}{3} = 2$$

$$6 \div \frac{1}{3} = 18$$

$$5^2 = 25$$

$$\sqrt{25} = 5$$

Example 1.5.2. The square root of a number that is not a perfect square is irrational. $\sqrt{9} = 3$ (rational), but $\sqrt{10}$ is irrational because 10 is not a perfect square.

Exercises 1.5

1.5.1. Compute each of the following.

(a) $9 + (-14)$

(b) $-6 - (-10)$

(c) $8 \times \frac{3}{4}$

(d) $9 \div \frac{1}{3}$

(e) 7^2

(f) $\sqrt{144}$

1.5.2. Identify which of the following are rational and which are irrational.

(a) $\sqrt{16}$

(b) $\sqrt{7}$

(c) $\sqrt{100}$

(d) $\sqrt{50}$

1.5.3. A student claims that $\sqrt{36} \times \sqrt{4} = \sqrt{36 \times 4}$. Verify whether this is true by computing both sides. Then try the same with $\sqrt{9} \times \sqrt{9}$.

1.6 Recursion

Many real-world processes are built from repeated steps rather than from a single calculation. A **recursion** describes such a process by specifying how each term is obtained from the one before it. In this course, two especially important patterns are additive recursion and multiplicative recursion.

Definition. *Additive recursion.* An **additive recursion** starts at a value a_0 and adds a fixed amount d at every step:

$$a_0, \quad a_0 + d, \quad a_0 + 2d, \quad a_0 + 3d, \quad \dots$$

The value at step n is $a_0 + nd$.

Definition. *Multiplicative recursion.* A **multiplicative recursion** starts at a value a_0 and multiplies by a fixed factor r at every step:

$$a_0, \quad a_0 \cdot r, \quad a_0 \cdot r^2, \quad a_0 \cdot r^3, \quad \dots$$

The value at step n is $a_0 \cdot r^n$.

Example 1.6.1. Additive recursion: start at 0, add 3 at every step.

$$0, \quad 3, \quad 6, \quad 9, \quad 12, \quad 15, \quad \dots$$

Each step increases the value by exactly 3. The value at step n is $3n$.

Example 1.6.2. Multiplicative recursion: start at 1, multiply by 2 at every step.

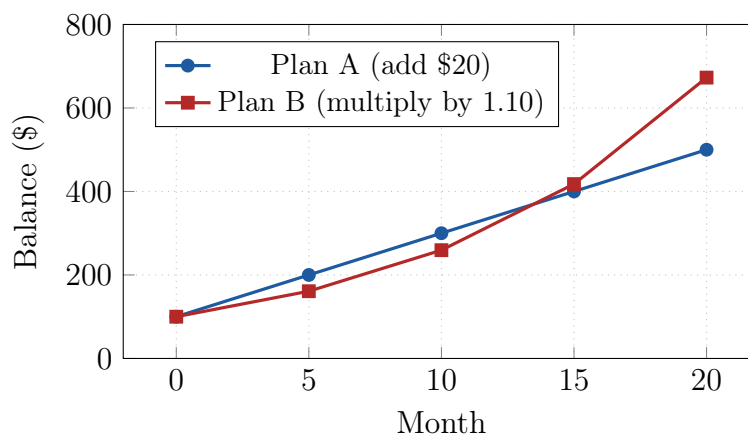
$$1, \quad 2, \quad 4, \quad 8, \quad 16, \quad 32, \quad \dots$$

Each step doubles the value. The value at step n is 2^n .

Example 1.6.3. Two savings plans each start with \$100. Plan A adds \$20 each month (additive). Plan B increases by 10% each month, that is, multiplies by 1.10 (multiplicative).

Month	Plan A	Plan B
0	\$100.00	\$100.00
1	\$120.00	\$110.00
2	\$140.00	\$121.00
3	\$160.00	\$133.10
4	\$180.00	\$146.41
5	\$200.00	\$161.05

Plan A leads through month 5. The multiplicative recursion of Plan B will eventually overtake any additive recursion. By month 12 Plan B reaches \$313.84 while Plan A reaches \$340.00 – still ahead – but by month 48, Plan B holds \$9,276.49 while Plan A holds only \$1,060.00.



Exercises 1.6

1.6.1. Write the first six terms of each recursion. Identify it as additive or multiplicative.

- (a) Start at 5, add 4 each step.
- (b) Start at 1, multiply by 3 each step.

(c) Start at 100, add -10 each step.

(d) Start at 2, multiply by $\frac{1}{2}$ each step.

1.6.2. A population of bacteria starts at 500 and doubles every hour.

(a) Write the population as a multiplicative recursion.

(b) Complete a table for hours 0 through 5.

(c) Write a formula for the population at hour n .

1.6.3. A car loses \$1,500 in value each year (additive recursion with a negative step). The car is initially worth \$18,000.

(a) Write the first six terms of the value sequence.

(b) After how many years does the car reach zero value under this model?

1.6.4. Two populations both start at 1,000. Population A grows by 200 each year. Population B grows by 15% each year. Without computing beyond year 5, predict which population will be larger in year 20. Explain your reasoning in two complete sentences.

Chapter 1 Review

The following exercises draw on material from the entire chapter.

6. Classify each number as $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{R}$ (smallest applicable).

(a) $\frac{8}{4}$

(b) -17

(c) $\sqrt{11}$

(d) $\frac{-6}{2}$

6. A diver starts at the surface (0 meters) and descends 15 meters, then ascends 8 meters, then descends 5 more meters. Write each move as an addition and give the final depth.

6. Determine whether $\frac{8}{14}$ and $\frac{12}{21}$ are equivalent. Simplify whichever fraction is not already in simplest form.

6. A recipe calls for $\frac{3}{4}$ cup of flour. If you want to make $\frac{2}{3}$ of the recipe, how much flour do you need? Show your calculation.

6. A colony of insects starts at 80 and triples every two weeks.
 - (a) Fill in the table for weeks 0, 2, 4, 6, 8.
 - (b) Write a formula for the population at time step n (where each step is two weeks).
 - (c) Is this an additive or multiplicative recursion?
6. A sequence begins 5, 5.5, 6, 6.5, 7, ...
 - (a) What type of recursion is this?
 - (b) Write the formula for the term at step n , starting from $n = 0$.
 - (c) What is the value at step 20?
6. Compute $\sqrt{64} + 3^2 - ((-4) \times 2)$. Show each step.
6. The number $\sqrt{2}$ appears in geometry as the length of the diagonal of a unit square. Explain in two complete sentences why this number is real but not rational.